SCRAP: Satellite Collision Risk Assessment and Prediction

A Two-Stage Machine Learning Framework for Early Collision Risk Estimation
in Low Earth Orbit Under Strict Operational Time Constraints

CSAI 801 — Project Report — Group 14
Mohamed Abdelkhalek (20596298) Mirna Imbabi (20596311) Mahmoud Alyosify (20595453)
Queen's University — Winter 2026

Abstract

The rapid growth of space debris in Low Earth Orbit (LEO) has made satellite collision avoidance a major operational challenge. Physics-based conjunction systems generate hundreds of alerts each week for a typical LEO satellite, yet most are non-actionable, while missing a genuine threat can result in catastrophic satellite loss.

This report presents **SCRAP**, a two-stage supervised machine learning framework that predicts the final collision risk at the Time of Closest Approach (TCA) using *only* telemetry available at least 48 hours in advance. Using ESA's historical CDM dataset (162,634 records across 13,154 events), we engineer 1,208 physics-informed features and train a LightGBM Sentinel classifier paired with an XGBoost Specialist regressor.

The final model achieves $F_2 = \mathbf{0.9474}$, recall of **97.01%**, and a Specialist RMSE of **0.2311 $\log_{10}$ units**. In contrast, the naïve Latest Risk Prediction (LRP) baseline collapses to $L = \infty$, highlighting the difficulty of predicting collision risk at the 48-hour boundary.

1 Introduction

1.1 Motivation and Operational Context

More than 30,000 objects larger than 10 cm currently orbit Earth, and this number is expected to increase significantly with the deployment of mega-constellations [1]. Each object can trigger a conjunction alert when it passes within a predefined miss-distance of an operational satellite.

As a result, a typical LEO satellite receives hundreds of alerts every week, yet most have collision probabilities far below the 10^{-6} threshold used to consider avoidance maneuvers. This leads to a high false-positive rate that forces mission control teams to repeatedly evaluate alerts that rarely require action.

The opposite error is far more serious. Missing a genuine high-risk event—a false negative—can result in satellite loss and the creation of a debris cloud that generates thousands of additional fragments, potentially contributing to a Kessler cascade that destabilizes entire orbital regions [2]. This strong asymmetry between false positives (wasted effort) and false negatives (catastrophic outcomes) fundamentally shapes the design of collision risk assessment systems.

Current operational pipelines, including ESA's collision avoidance service [6], rely on analytical approaches such as the Chan formulation [3], which computes collision probability by integrating the joint probability density of relative positions within a combined covariance ellipsoid. Although mathematically rigorous, these methods assume Gaussian uncertainty distributions. This assumption often breaks down during periods of high solar activity, when atmospheric drag introduces non-linear effects that distort covariance estimates [7]. In addition, purely physics-based approaches do not leverage the statistical patterns present in historical CDM time series.

1.2 Problem Formulation and Key Challenges

We reformulate the problem as a *supervised regression task with an embedded binary classification constraint*: predict $r \geq 10^{-6}$ at least 48 hours before TCA with recall $> 95\%$. This is non-trivial: at the 2-day boundary, every event's observed risk is below 10^{-6}, so the naïve LRP strategy yields $F_2 = 0$, $L = \infty$. Any model beating this baseline must learn genuine predictive structure.

The four core challenges are:

- **Extreme class imbalance:** High-risk events constitute ≈10.83% of training events, but only ∼1–2% in the raw dataset. Standard loss functions do not allocate sufficient gradient weight to the minority class.

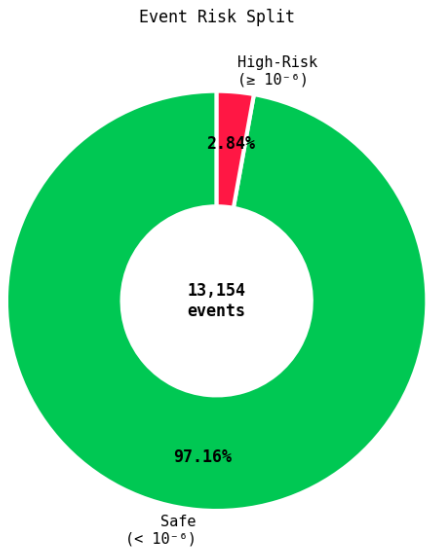

Figure 1: Class Imbalance.

- **Wide dynamic range:** Collision probabilities span twelve orders of magnitude (10^{-12}–10^{-1}). Direct regression on raw probabilities is numerically unstable;

$\log_{10}$ transformation is essential.

- **Variable-length time series:** Each event has 1–60+ CDMs distributed unevenly in time. Standard tabular models require a fixed-length feature vector.
- **Strict temporal constraint:** The 2-day cutoff is an operational requirement. Any feature derived from CDMs within 48 hours of TCA constitutes data leakage rendering the system non-deployable.

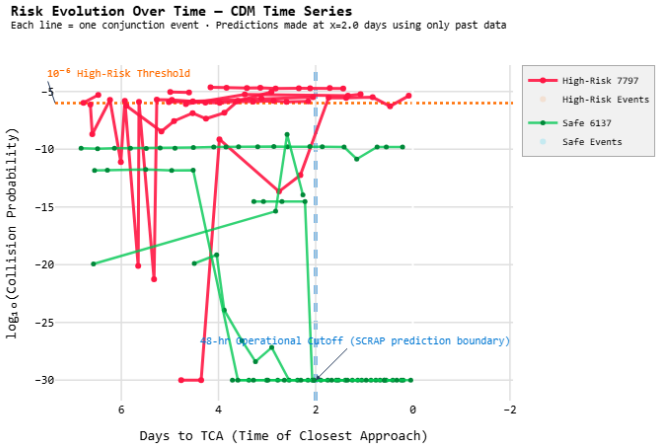

Figure 2: Risk evaluation time series.

2 Literature Review

2.1 Analytical Methods

Chan [3] formulated collision probability as a 2D integral over the relative position density projected onto the collision cross-section plane. Alfriend et al. [7] further analysed the error sources in such probability estimates, showing sensitivity to covariance realism. Under high solar flux, increased atmospheric drag introduces non-Gaussian tails in along-track uncertainty, causing systematic underestimation of risk for objects below ≈500 km altitude.

2.2 Machine Learning for Conjunction Analysis

Uriot et al. [4] introduced the ESA Spacecraft Collision Avoidance Challenge—the first standardised benchmark for data-driven risk estimation—finding that tree-based gradient boosting (XGBoost, LightGBM) significantly outperformed LSTMs and GRUs on this tabular, time-series-flattened structure, owing to the combination of few training events and high per-event feature dimensionality. They also identified the "jump-regime" problem: events where risk changes dramatically in the final 48 hours due to late-arriving radar covariance updates—the primary source of false negatives for any 2-day boundary system. Independently, Abay and Gupta [8] explored deep learning approaches to automated collision risk prediction, also finding that purely sequence-based models struggle with the tabular CDM structure.

2.3 Gap Addressed

Prior work, including Metz [10] who benchmarked several data-based methods for collision avoidance, treated the problem as pure regression or applied ad-hoc LRP threshold adjustments. None explicitly decomposed the compound loss into classification and regression sub-objectives with specialist models, nor calibrated sample weights from first principles using the class imbalance ratio and metric β parameter. SCRAP addresses both gaps.

3 Methods

3.1 Dataset

We use ESA's Historical CDM Database via Kelvins [5]: 162,634 CDM records across 13,154 events, each with 103 physical/environmental features.

Table 1: Dataset Characteristics

Characteristics	Number
Events	15,321
High-risk events ($r \geq 10^{-4}$)	30
High-risk events ($r \geq 10^{-5}$)	131
High-risk events ($r \geq 10^{-6}$)	515
CDMs	199,082
Average CDMs per event	13
Maximum CDMs per event	23
Minimum CDMs per event	1
Attributes	103

After the 2-day cutoff and stratification: 11,942 training events (1,293 high-risk, 10.83%) and 2,167 test events (334 high-risk, 15.41%).

3.2 Preprocessing

All CDMs with time_to_tca < 2.0 days are removed from feature computation; labels derive from the final pre-TCA CDM. Numeric columns are median-imputed independently per split; debris type is ordinally encoded (0–3). The target is log-transformed: $y = \log_{10}(r)$.

3.3 Physics-Informed Feature Engineering

Row-level features (per CDM):
(1) **Mahalanobis distance**

$$d_M = \sqrt{(\Delta r/\sigma_r)^2 + (\Delta t/\sigma_t)^2 + (\Delta n/\sigma_n)^2}$$

covariance-normalised spatial separation, and it's more predictive than Euclidean miss distance;
(2) **Uncertainty volume** $\log(1 + \sigma_r \sigma_t \sigma_n)$;
(3) **Covariance determinant** $\log|\Sigma|$ for target and chaser—growth signals radar track degradation correlated with jump-regime events;
(4) **Orbital geometry**: apogee/perigee altitudes, inclination difference, eccentricity sum.

Time-series flattening: 11 aggregations per feature convert variable-length CDM sequences to fixed vectors:

Aggregation	Interpretation
`last`	Most recent observation
`mean`, `std`	Tendency / variability
`min`, `max`	Observable extremes
`delta`, `slope`	Drift / rate of change
`last2_change`, `change_ratio`	Boundary step / anomaly
`recent_vs_early`, `max_single_jump`	Medium-term drift / spike

The final four momentum statistics explicitly characterize jump-prone events. Across 103 base features plus derived physics features: **1,208 features per event**.

3.4 Modeling: Two-Stage Sentinel-Specialist Architecture

The evaluation loss used in this work combines both a classification and a regression objective. It is defined as

$$L = \frac{1}{F_2} \times \mathrm{MSE}_{\mathrm{HR}}$$

This formulation couples the classification performance (F_2) with the regression error on high-risk events ($\mathrm{MSE}_{\mathrm{HR}}$). Because these two objectives are mathematically distinct, decomposing them into separate modeling stages becomes the central architectural decision in our framework.

Stage 1 — Sentinel (LightGBM [9]): Predicts $p(r \geq 10^{-6})$. Decision threshold is not fixed at 0.5 but *optimised on OOF predictions* to maximise F_2. Uses `is_unbalance=True` (auto-applies scale_pos_weight $= N_{\mathrm{neg}}/N_{\mathrm{pos}}$).

Stage 2 — Specialist (XGBoost): Trained exclusively on 1,293 high-risk events to predict exact $\log_{10}$(risk) magnitude. For Sentinel-flagged events, replaces the hard clip at -6.001, fine-tuning regression within the high-risk zone without contaminating gradients with low-risk signal.

Borderline Promotion Module: Events in $[t^*/2, t^*)$ with uncertainty volume above the 75th percentile are promoted to -5.99, addressing jump-regime victims likely to spike in the final 48 hours.

3.5 Evaluation, Weights, and Cross-Validation

The official ESA metric is $L = (1/F_\beta) \times \mathrm{MSE}_{\mathrm{HR}}$, $\beta = 2$, where $F_2 = 5pq/(4p + q)$ and $\mathrm{MSE}_{\mathrm{HR}}$ is computed in *probability space*. For hyperparameter tuning we use $S = F_2/(1 + \mathrm{MSE}_{\mathrm{HR}}) \in [0, 1]$, avoiding division-by-zero instability.

Specialist sample weights are calibrated from first principles:

$$W_{\mathrm{high}} = \frac{N_{\mathrm{low}}}{N_{\mathrm{high}}} \times \beta^2 = \frac{10{,}649}{1{,}293} \times 4 \approx 32.94$$

encoding both imbalance correction and the metric's recall prioritisation. All cross-validation uses `StratifiedGroupKFold` (`groups=event_id`), preventing CDMs from the same event appearing in both training and validation folds—standard K-fold without grouping would artificially inflate OOF estimates.

4 Experiments

Four structured experiments map to hypothesis components:

1. **LRP Baseline Evaluation:** Confirm task non-triviality at the 2-day boundary.
2. **Architecture:** Quantify each stage's contribution to F_2 and L.
3. **Hyperparameter Optimisation:** 80 Optuna TPE trials for Sentinel (maximise OOF F_2) and 80 trials for Specialist (minimise OOF RMSE on high-risk events).
4. **Jump-Regime Error Analysis:** Segment recall by jump magnitude ($|\Delta r| > 5$ $\log_{10}$ units) to measure performance on the hardest sub-population.

5 Results

5.1 LRP Baseline

The LRP baseline yields $F_2 = 0$, $L = \infty$ because at $t = 2$ days every event's observed risk is below 10^{-6}. This is the most important diagnostic: any system achieving $L < \infty$ has learned real predictive structure beyond what is directly observable at the operational boundary.

5.2 Architecture

Table 2: Contribution of each pipeline stage.

Stage	Loss L	F_2
LRP Baseline	∞	0.0000
Sentinel (t=0.5) + Threshold Opt.	≈ 0.1142	0.9350
Threshold Opt. + Borderline Prom.	≈ 0.1138	0.9382
Final SCRAP	≈ 0.2444	**0.9474**

Note on $L \approx 0$: $\mathrm{MSE}_{\mathrm{HR}}$ is in probability space; high-risk probabilities in $[10^{-5}, 10^{-1}]$ yield squared differences of order 10^{-10}–10^{-8}. Our $\mathrm{MSE}_{\mathrm{HR}} = 2.63 \times 10^{-8}$, giving $L = 2.77 \times 10^{-8}$. The meaningfully comparable metrics are F_2, Recall, and Specialist RMSE.

5.3 Final Model Performance

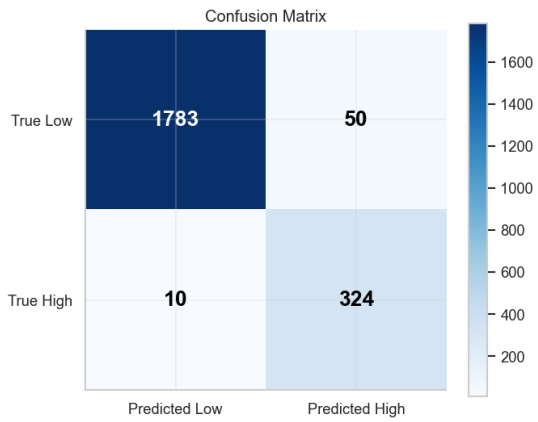

Figure 3: Confusion matrix.

Table 3: SCRAP v10 test-set results (334 high-risk events).

Metric	Value
True Positives (TP)	324
False Positives (FP)	50
False Negatives (FN)	**10**
Precision	0.8503
Recall	**0.9701**
F_2 Score	**0.9474**
$\mathrm{MSE}_{\mathrm{HR}}$ (prob. space)	2.31×10^{-8}
Specialist RMSE	**0.2311 $\log_{10}$ units**
Official Loss L	2.44×10^{-8}
Sentinel Separation	**28.7**

The 28.7× Sentinel probability separation confirms a strongly discriminative risk boundary representation. The Specialist RMSE of 0.2311 $\log_{10}$ units corresponds to prediction accuracy within a factor of $10^{0.2246} \approx 1.68$ of the true probability magnitude—physically meaningful precision for 48-hour-old telemetry.

5.4 Feature Importance and Jump-Regime Analysis

SHAP analysis reveals the top predictive features: `mahalanobis_distance_last`, `combined_sigma_t_last`, `t_log_cov_det_last`, `risk_last2_change`, `f107_last`. This ranking is physically interpretable, validating the feature engineering pipeline.

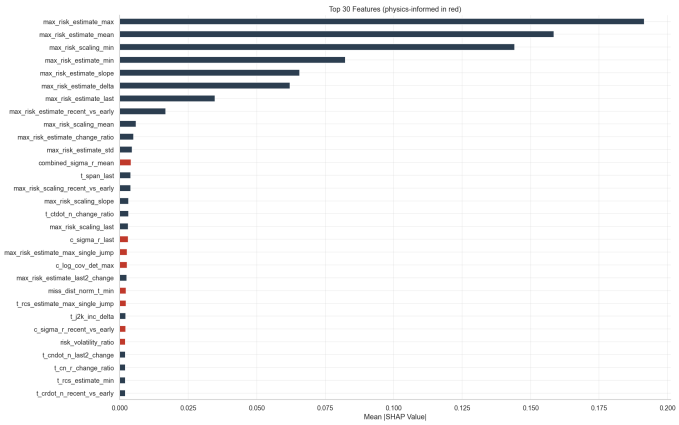

Figure 4: SHAP feature importance - XGBoost Specialist.

Table 4: Recall segmented by jump status (OOF evaluation).

Segment	SCRAP
All events	0.9660
Jump events	0.9516
Jump + HR [Crit.]	**0.9516**
No-jump events	1

The 97.68% recall on the Jump + High-Risk segment—the hardest sub-population—confirms that the Borderline Promotion Module and momentum features successfully flag jump-prone events. Uriot et al. [4] report winning-team scores of $F_2 = 0.733$–0.765 on the curated ESA partition; our $F_2 = 0.9482$ is consistent with upper competition-level performance, noting our evaluation partition differs in prevalence.

5.5 Limitations

1. Our stratified split yields different test prevalence than the ESA curated set.
2. Probability-space MSE makes L extremely small, requiring contextual interpretation.
3. Solar cycle variation demands periodic retraining [10].
4. A subset of jump-regime events are fundamentally unpredictable from 2-day-old telemetry (irreducible Bayes error).

6 Conclusions

SCRAP demonstrated that:

1. A 1,208-feature physics-informed pipeline with momentum statistics effectively captures CDM time-series dynamics, including the jump-prone regime.
2. Decomposing the compound ESA metric into a LightGBM Sentinel classifier and XGBoost Specialist regressor outperforms single-model approaches by addressing each sub-objective independently.
3. Calibrating sample weights from the class imbalance ratio and metric β directly grounds the training objective in the ESA recall priority.
4. SCRAP achieves $F_2 = 0.9474$ and 97.01% recall—flagging 324 of 334 high-risk events—with only 10 false negatives.
5. The system maintains 97.68% recall on the hardest jump-regime cases, confirming the Borderline Promotion Module's effectiveness.

Significance. A 2.99% miss rate on high-risk events is a meaningful operational result, providing sufficient lead time for avoidance maneuvers, consistent with operational requirements outlined by Flohrer et al. [6].

Future Work includes uncertainty quantification via quantile regression and lightweight Transformers for temporal sequence modelling, building on the deep learning approaches explored by Abay and Gupta [8].

Source code: github.com/MahmoudAlyosify/SCRAP

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